

Developing Pharmaceutically Relevant Pd-Catalyzed C–N Coupling Reactivity Models Leveraging High-Throughput Experimentation

Seung Kyun Ha,[#] Dipannita Kalyani,^{*,#} Michael S. West,[#] Jessica Xu,[#] Yu-hong Lam, Thomas Struble, Spencer Dreher, Shane W. Krska, Stephen L. Buchwald, and Klavs F. Jensen^{*}Cite This: *J. Am. Chem. Soc.* 2025, 147, 19602–19613

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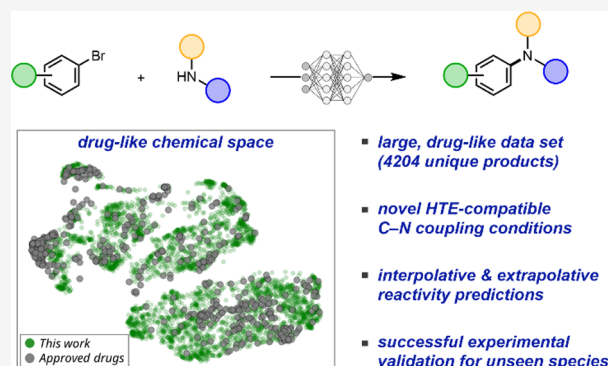


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ABSTRACT: This manuscript presents machine learning models for Pd-catalyzed C–N couplings constructed using a large, pharmaceutically relevant, structurally diverse dataset (4204 unique products) generated *de novo* using high-throughput experimentation. The dataset generation was enabled by the discovery of novel nanomole scale compatible automation friendly C–N coupling reaction conditions using LiOTMS as the base. The large dataset enabled the systematic evaluation of model performance using five different data-splitting strategies that were carefully designed to assess the models' ability to both interpolate and extrapolate. The models exhibit high predictive performance across all splits as gauged by standard metrics. In addition, the models predicted with high accuracy the outcome of validation libraries that were outside the scope of the training set. Employing these models in the context of medicinal chemistry campaigns should result in significant enrichment of successful C–N couplings.



INTRODUCTION

The merger of high-throughput experimentation (HTE) and data science presents an enormous opportunity to both accelerate and inspire innovations in synthetic chemistry. Contemporary HTE enables the rapid and efficient exploration of diverse chemical spaces and reaction conditions while ensuring consistent data capture.^{1–6} Similarly, developments in machine learning (ML) have enabled the distillation of large and complex datasets into predictive models capable of generalizing patterns in the data.^{4,7–14} Despite these advances, efforts to merge HTE with ML remain largely limited to a few reported datasets with limited structural diversity.^{15–21}

An important application of HTE is optimization of reaction conditions for single or multiple target compounds.^{20,22,23} While identifying reaction conditions using multiparameter optimization is fruitful, it is time and resource intensive, especially in the context of medicinal chemistry campaigns for which large libraries of target analogues are often required. An efficient and forward-looking approach entails the creation of predictive reactivity models to prospectively assess reaction performance *in silico*, which would enable researchers to identify low- or high-yielding reactions, thereby focusing time-intensive screening efforts only on targets predicted to be low-yielding. Despite their potential impact, predictive models for drug-discovery relevant transformations that encompass the structural complexity and diversity encountered in pharmaceutical targets remain sparse.²⁴ To this end, we embarked on an effort to build predictive models

for palladium-catalyzed C–N cross-coupling, a workhorse reaction in medicinal chemistry, by leveraging the state-of-the-art HTE and ML methods.

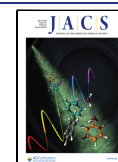
A survey of prior work reveals that efforts to acquire and model the reaction outcome of C–N coupling data fall under two distinct categories, employing historical (**Strategy I**) or HTE-derived (**Strategy II**) datasets (Figure 1A). Pd-catalyzed C–N coupling data have been extracted from historical reaction datasets such as patent databases and Reaxys^{8,9,25–28} as well as Electronic Laboratory Notebooks.²⁹ Predictive reactivity models built for either reaction-specific²⁹ or general-purpose³⁰ performances using historical datasets (**Strategy I**) exhibit relatively poor performance [as evidenced by low correlation coefficient ($R^2 \sim 0.2$)], in part due to significant heterogeneity in the data quality which can be very challenging to curate systematically. Furthermore, while these historical datasets are relatively large, they are often biased toward high-yielding experiments, whereas prediction models benefit from the inclusion of a distribution of successful and failed experiments.^{29–32} Additionally, these historical datasets might not

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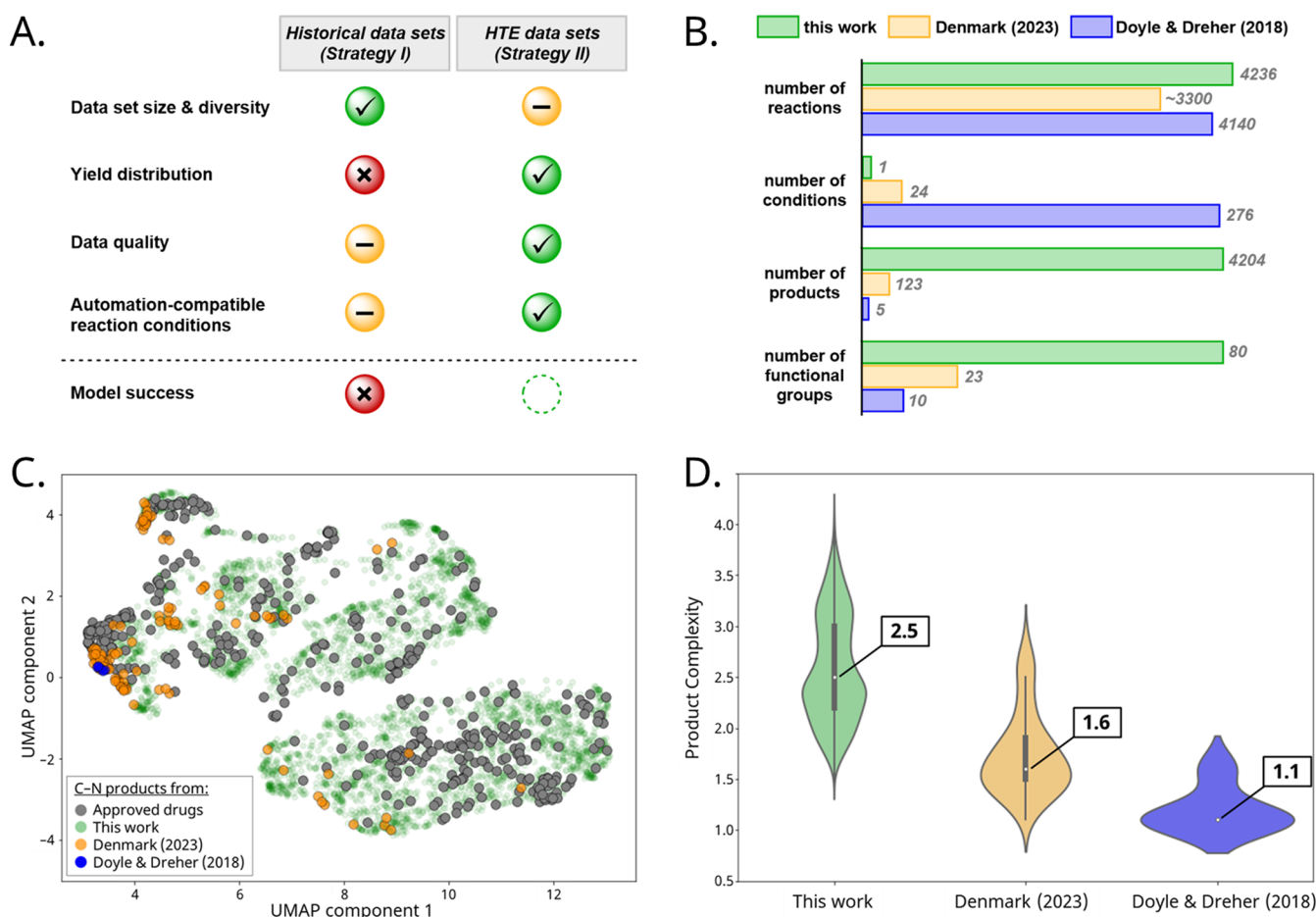


Figure 1. (A) Comparison of historical and HTE datasets. (B) Comparison of this work with previous reports. (C) UMAP visualization of the HTE datasets in this work, previous reports, and approved drugs from DrugBank containing a C–N bond. UMAP visualization constructed using 14 physical chemical drug-related properties calculated using RDKit (see SI for details). (D) Complexity score of HTE dataset products in this work and previous reports.

contain the state-of-the-art HTE-amenable conditions for library synthesis as adoption of modern methods often lags behind the current literature.^{26,28}

Yield-modeling efforts on the few reported datasets demonstrate the potential advantages of **Strategy II**. These include the generation of large, homogeneous high-quality datasets without bias toward high-yielding reactions.^{15,16,21,33} Doyle, Dreher, and co-workers reported the first large HTE data for C–N coupling modeling efforts (~4K reactions) albeit with very few (5) structurally distinct cross-coupled products (Figure 1B).¹⁶ The goal of this report was to predict the performance of C–N couplings in the presence of isoxazole additives which yielded predictive models with $R^2 > 0.9$.^{11,16,29,30,34–38} However, the objective of this work was largely related to a mechanistic question, and thus, the extrapolative ability of these models to predict the performance of cross-couplings beyond those in the training set was not investigated. More relevant to this manuscript is the recent report by the Denmark group on predictive models for C–N couplings using a HTE dataset comprised of ~123 unique cross-coupled products and 24 distinct reaction conditions.²¹ The goal of this work was to predict the optimal reaction conditions and performance for C–N couplings of a broad range of aryl halides and amines. While it is a step forward from the Doyle & Dreher report stated above with regards to the structural diversity of the dataset, the overlap between the cross-coupled products versus the drug-like

chemical space³⁹ is narrow (Figure 1C, orange circles) lending uncertainty to the applicability of these models for pharmaceutically relevant complex C–N couplings.

Herein, we present a complementary approach to previous reports by building predictive models using a large structurally diverse HTE dataset comprising one reaction condition and 4204 unique C–N coupling products (Figure 1C). The higher average complexity score (Figure 1D),⁴⁰ the greater functional group diversity (Figure 1B), and the significant overlap of the cross-coupled products with the drug-like chemical space (Figure 1C) should manifest in enhanced confidence for pharmaceutical relevance of the resulting predictive models.

The structural diversity in the dataset was enabled by leveraging the Merck & Co., Inc., Rahway, NJ, USA Building Block Collection (MBBC) which comprises >20,000 amines and >7000 aryl halides, theoretically allowing access to hundreds of millions of products. Our initial efforts focused on C–N coupling using secondary amines, thereby reducing this virtual chemical space to ~15 M products using ~5000 aryl bromides and ~3000 secondary amines (Figure 2). The generation of the large dataset was enabled by the discovery of state-of-the-art automation friendly, nanomole scale chemistry compatible C–N coupling conditions. It is imperative that reaction models be constructed using the most robust HTE-amenable reaction conditions to maximize applicability across a broad range of pharmaceutically relevant chemical space. Careful consideration

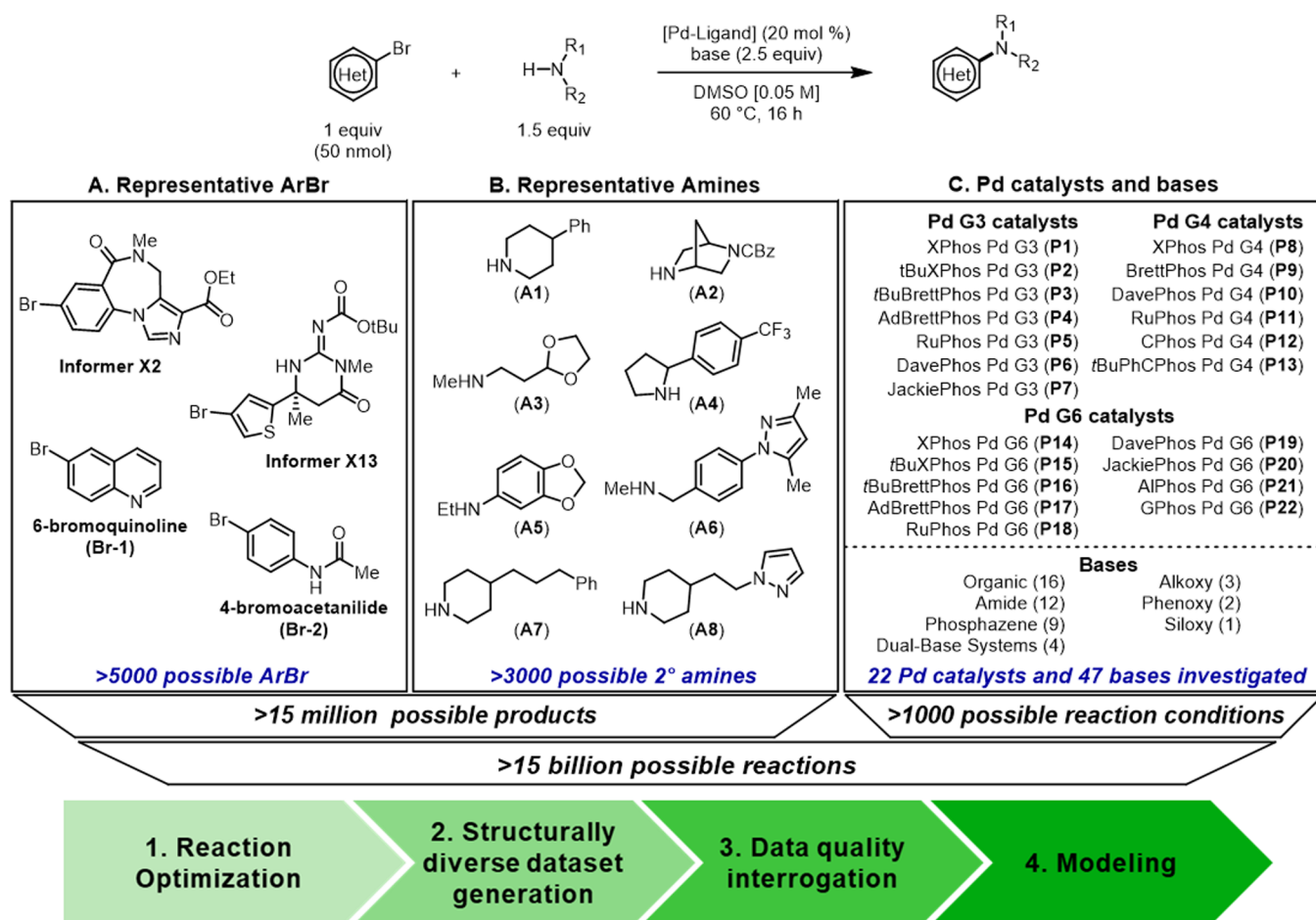


Figure 2. Optimization workflow with examples of aryl halides, amines, Pd precatalysts, and bases investigated.

was given to both the systematic assessment of data quality and the extrapolative performance of the models.

RESULTS AND DISCUSSION

General Considerations for the Generation of Large, Diverse Datasets. As mentioned above, our efforts aimed to generate predictive models for the Pd-catalyzed C–N coupling of secondary amines. Several experimental considerations were important for the creation of requisite datasets, including (1) the identification of appropriate nanomole scale amenable, automation friendly C–N coupling conditions to facilitate running large numbers of experiments with high levels of control, (2) selection and generation of a structurally diverse molecular dataset mirroring the complexity present in pharmaceuticals, and (3) systematic interrogation of data quality. The following sections detail our workflow, addressing each of these considerations.

Identification of Nanomole Scale Compatible Pharmaceutically Relevant Reaction Conditions. To date, efforts on C–N couplings using nanomole scale (0.1 μ mol) chemistry have employed tBuXPhos Pd G3 catalyst and P₂Et as the base.¹⁸ The performance of these reaction conditions for Pd-catalyzed C–N coupling have been benchmarked for the coupling of Informer halides with piperidine at microscale (2.5 μ mol).¹⁸ The Informer halides are a standardized set of 18 complex halides that comprise a breadth of structural features, functional groups, and physiochemical properties commonly encountered in pharmaceutical targets (see Figure S7 for

structures). Hence, the Informer set has been widely used to assess the relevance of synthetic methods for the functionalization of drug-like substrates.^{41–43} These studies revealed significant opportunity for the identification of more robust reaction conditions since only 4/18 Informer halides were transformed to the desired products with $\geq 20\%$ product LCAP (liquid chromatography area percent).⁴⁴ Hence, in a series of preliminary studies, we aimed to identify the catalyst and base system that enabled the broadest scope with respect to the coupling of drug-like aryl halides with secondary amines. A product LCAP threshold of 20% was used to assess the degree of product formation.⁴⁵ As shown in Figure 2, we investigated the use of 22 distinct Pd catalysts generated through combinations of 12 biaryl ligands and 3 different generations of precatalysts (G3, G4, or G6). Additionally, 47 distinct bases were used for these studies. The precatalysts and bases were largely chosen to maximize structural diversity and ensure solubility in the nanochemistry compatible reaction solvent (DMSO). Theoretically, >1000 reagent combinations exist among these precatalysts and bases. Together with the >15 million possible products, the total number of possible reactions for a full factorial evaluation of all reaction parameters and substrates exceeds 15 billion. As detailed below, our nanoscale experimental workflow enabled us to identify suitable reaction conditions for the generation of large datasets using a subset of this vast possibility of reaction permutations.

In a 1536-well plate, the coupling of 3 aryl bromides [6-bromoquinoline (**Br-1**), 4-bromoacetanilide (**Br-2**), and

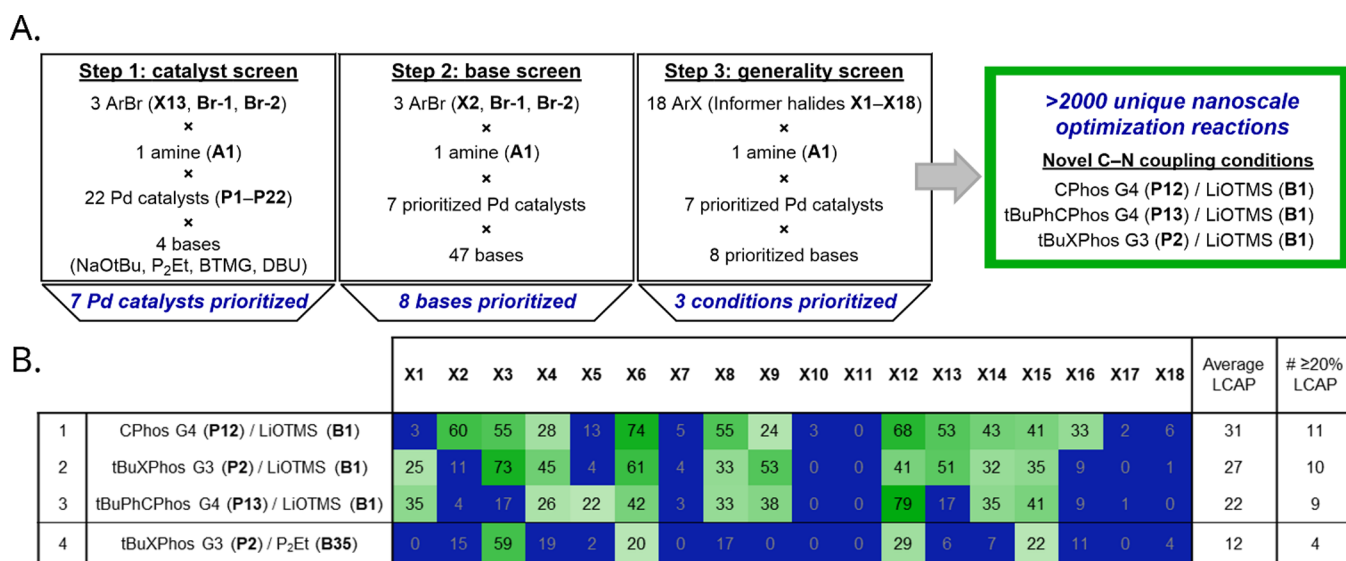


Figure 3. (A) Nanoscale reaction optimization and elucidation of best conditions. For each step, aryl halide (1.0 equiv), amine (1.5 equiv), Pd precatalyst (20 mol %), and base (1.5 equiv) was used (see SI for details). (B) Product LCAPs for step 3; see Figure S7 for structures of Informer halides.

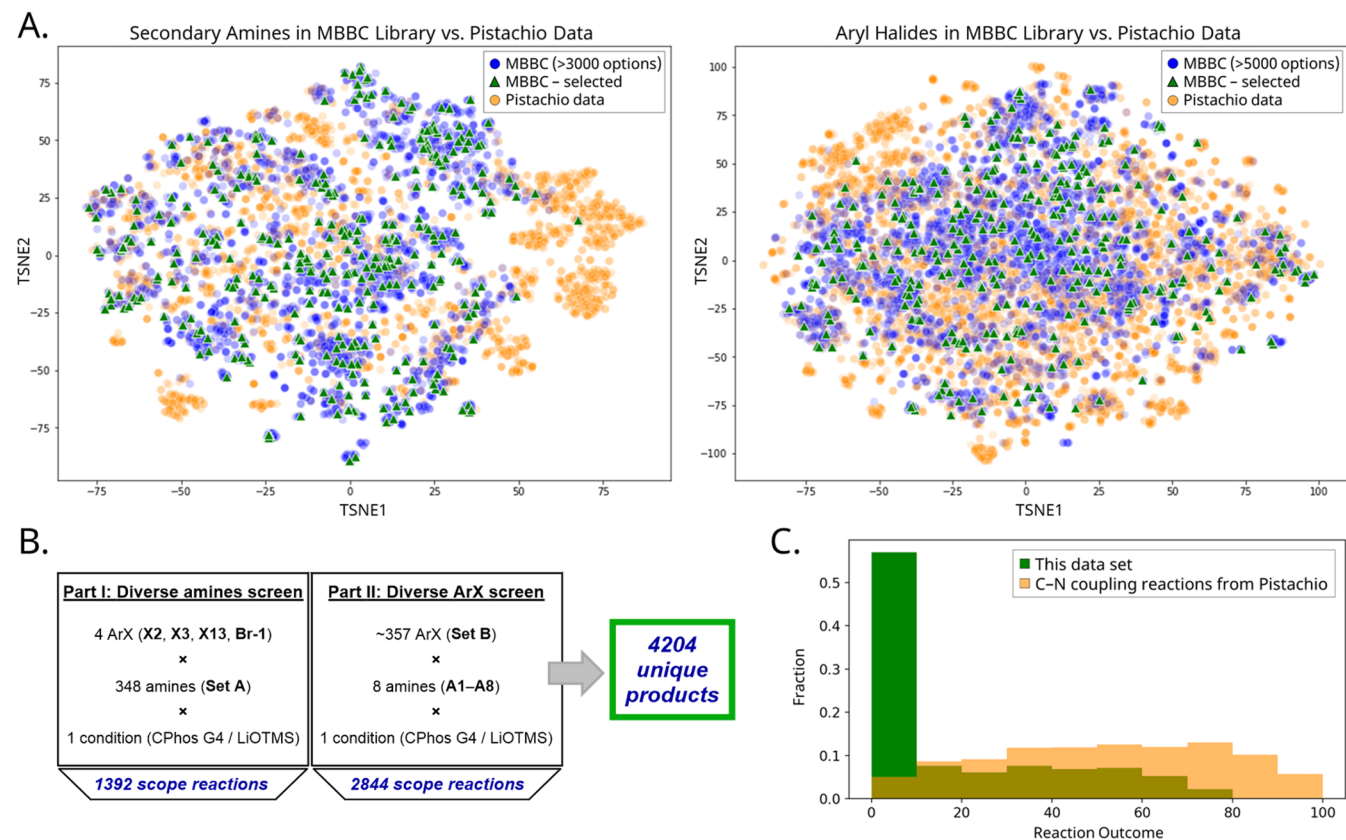


Figure 4. Summary of the constructed dataset. (A) Amine and aryl halide chemical space visualizations: t-distributed stochastic neighbor embedding (t-SNE) plots of MBBC pools of suitable amines (left) and aryl halides (right) with chosen molecules marked. The plots visualize 2048-bit, radius-2 Morgan fingerprints of molecules in the 2D space. (B) Workflow for generation of structurally diverse datasets. In **Part I**, 4 aryl halides were coupled with 348 amines resulting in 1392 scope reactions. In **Part II**, 357 aryl halides were coupled with 4 amines (1428 reactions) and only 354 aryl halides (due to availability) were coupled with the remaining 4 amines (1416 reactions), resulting in a total of 2844 scope reactions. This results in a combined total of 4236 scope reactions consisting of 4204 unique products (without duplicates). Due to the presence of some amines as salts in **Set A**, 2.5 equiv base was used for **Part I** and **Part II**. (C) Yield distributions for the generated dataset versus an isolated set of C–N coupling reactions from the Pistachio patent dataset. Reaction outcome = product % yield for the Pistachio dataset and product LCAP for the HTE dataset.

Informer halide (X13)] with 4-phenylpiperidine (A1) was conducted using 22 Pd precatalysts and 4 bases (Figure 3A, step

1). To identify conditions that are amenable to couplings with pharmaceutically relevant halides, X13 was intentionally

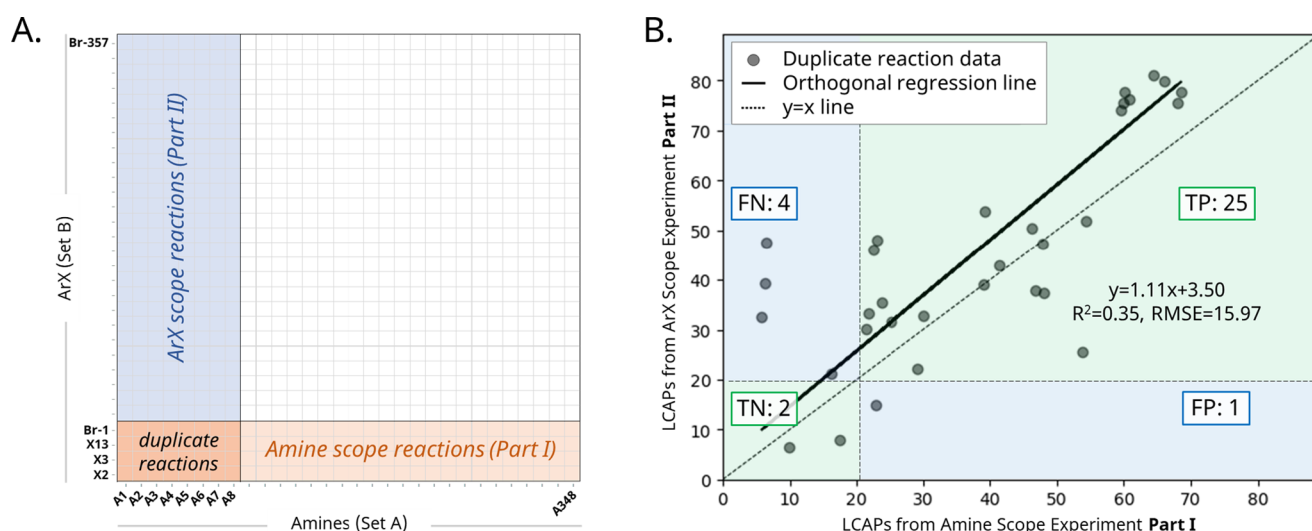


Figure 5. Classification analysis for assessment of reaction reproducibility. (A) Schematic illustrating duplicate reactions between amine and aryl halide scope studies. (B) Reproducibility study using duplicate scope reactions. TP = true positive; FP = false positive; TN = true negative; FN = false negative.

included based on prior experience as a particularly challenging halide for C–N couplings. This screen led to the prioritization of 7 precatalysts for further study. These 7 precatalysts were used in combination with 47 bases for the coupling of **X2**, **Br-1**, and **Br-2** with 4-phenylpiperidine (**A1**) to identify 8 bases that led to the highest product LCAP (Figure 3A, step 2).⁴⁶ To assess the robustness and generality of the 56 conditions (7 precatalysts $\times$ 8 bases) resulting from the prioritized precatalysts and bases, the reactions of 4-phenylpiperidine with 18 complex aryl halides (Informer halides **X1**–**X18**) were assessed (Figure 3A, step 3). As shown in Figure 3A, three sets of conditions [CPhos Pd G4 (**P12**) with LiOTMS (**B1**), (*t*Bu)PhCPhos Pd G4 (**P13**) with LiOTMS (**B1**), and *t*BuXPhos Pd G3 (**P2**) with LiOTMS (**B1**)] demonstrated a more general scope than conditions previously used [*t*BuXPhos Pd G3 precatalyst (**P2**) with P_2Et as the base].¹⁸ The success rates and average product LCAPs with the newly identified conditions (Figure 3B, entries 1–3) are the best reported to date for nanoscale amenable Pd-catalyzed C–N couplings. These conditions led to ~ 2 – 3 -fold increase in the number of Informer halides giving products with $\geq 20\%$ LCAP versus the previous method using *t*BuXPhos Pd G3 (**P2**) with P_2Et (**B35**) as the base. Ultimately, we utilized a single set of conditions, **P12** and **B1**, for all subsequent studies. Importantly, the use of LiOTMS as a base for C–N coupling was unprecedented prior to this report, emphasizing the importance of large unbiased multiparameter optimizations for the discovery of novel reaction conditions.⁴⁷

Selection of Aryl Halides and Amines. Having determined the appropriate automation friendly reaction conditions set the stage to generate the requisite large HTE datasets for modeling. The first step was to select a diverse array of pharmaceutically relevant aryl halides and amines. Of the aryl bromides (>5000) and amines (>3000) that met the functional group filtration criteria (e.g., highly reactive electrophiles such as acyl chlorides were eliminated), 347 amines and 342 aryl bromides were randomly selected (Figure 4A). These selected ArBr and amines span the chemical space well and represent the diversity of building blocks in MBBC and the Pistachio patent database.^{48,49} While we resorted to using a random selection process, clustering of the MBBC chemical space was evaluated and showed that the selected coupling partners (based on

random selection) span 191/250 and 180/250 clusters for amines and ArBr, respectively (see Figures S61 and S62).

Generating a Large, Diverse HTE Dataset. The next step was to generate the HTE dataset using selected amines and ArBr. Generating a full-factorial dataset (n^2 space) using all selected halides and amines would require ~ 120 K experiments, which, while achievable, is prohibitively time and resource intensive even with state-of-the-art HTE data analysis techniques. Considering this practical limitation, the ability to generate predictive models using a fraction of the n^2 space is highly desirable.¹⁷ Herein, we demonstrate that our dataset, which comprises a small fraction (4204 unique products) of the full factorial space, was sufficient to predict well on the n^2 space and beyond (*vide infra*). The generation of the large datasets for modeling was conducted in two parts (**Part I** and **Part II**, Figure 4B). **Part I** involved the coupling of 348 secondary amines [Set A: 4-phenylpiperidine (used for reaction optimization) + the 347 randomly selected amines] with 4 aryl bromides. Since aryl halides from the Informer set were used to identify the optimal C–N coupling conditions (Figure 3), three of the four bromides used for **Part I** (**X2**, **X3**, and **X13**) were also selected from that set. The fourth halide was the simplest partner—6-bromoquinoline (**Br-1**).

Part II entailed the coupling of the 357 aryl halides (Set B: 15 Informer halides + the 342 randomly selected aryl bromides)⁵⁰ with 8 secondary amines (**A1**–**A8**, for amine structures see Figure 2). The amines for **Part II** were chosen using the results from **Part I** to maximize structural diversity and varied reactivity (see the SI for details). Importantly, as mentioned in the introduction, the chemical space covered by approved drugs intersects well with the structural diversity of the products generated in **Part I** and **Part II**, affirming the relevance of the generated HTE data for applications in drug discovery (Figure 1C). Furthermore, the product yield distribution from the HTE dataset is representative of realistic reaction performance with a mode of 0% product LCAP versus the literature-derived datasets (e.g., Pistachio) that largely report successful reactions (Figure 4C).

Data Quality Interrogation. As emphasized in the literature, the availability of high-quality systematic data is imperative for building reliable and robust predictive models.³¹

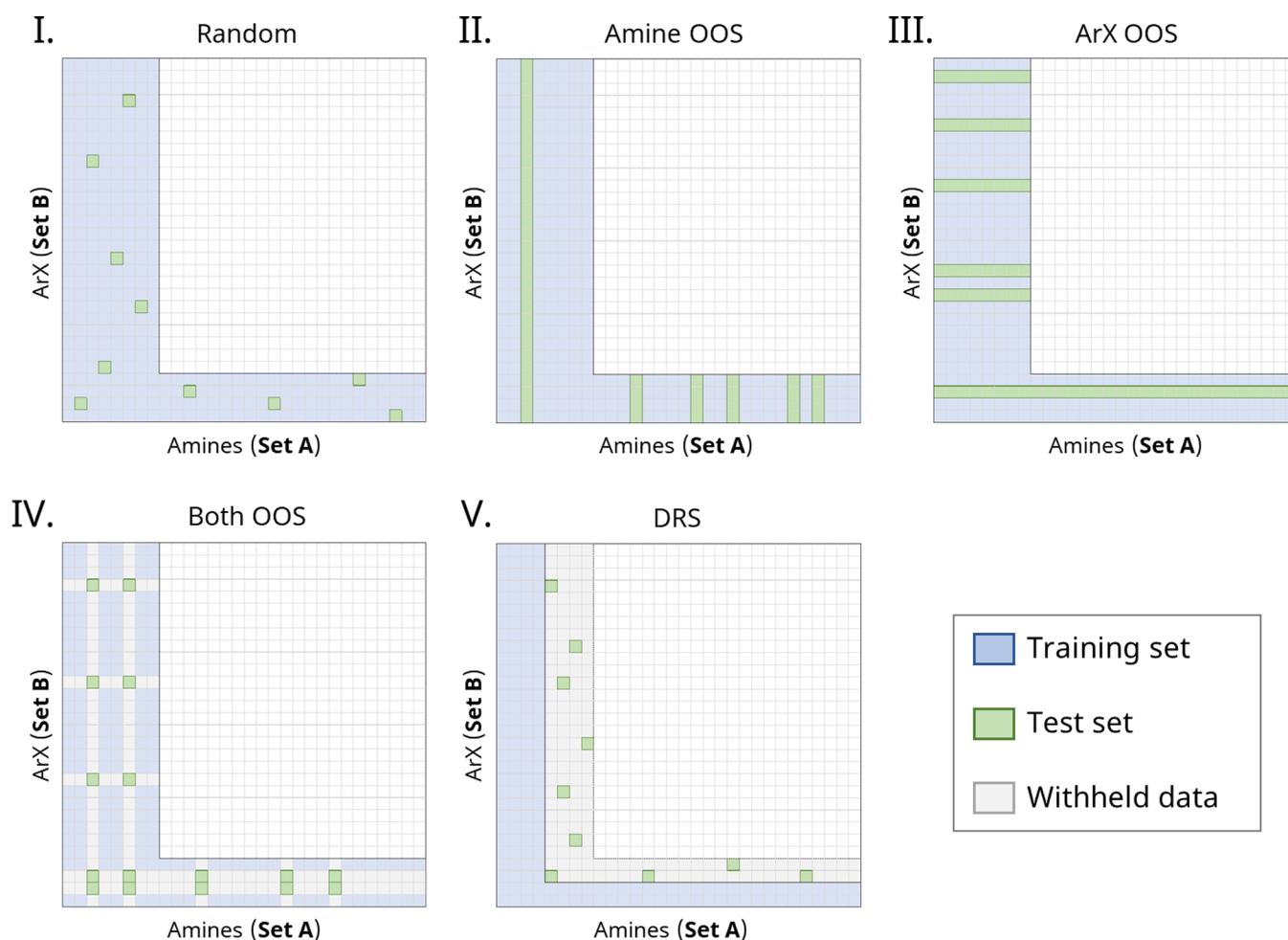


Figure 6. Dataset splitting strategies with training data in light blue, test data in light green, withheld data in light gray, and uncollected data in white for different data splits (OOS = out-of-scope). Also see Figures S63–S70 and Table S1.

Hence, careful consideration of potential experimental sources of error was deemed important.⁵¹ As detailed below, two major sources of experimental error were considered: substrate quality and experimental reproducibility.

Assessment of Substrate Quality. Due to the large number of substrates involved in this study, it was deemed impractical to individually confirm the identity of each aryl halide and amine. Instead, proxy reactions were performed. For a subset of amines (62/347), an independent sample from an external vendor (Enamine) was acquired, and the set of previously performed C–N coupling reactions in Part I (Figure 4B) was repeated using the new samples. Based on these studies, 90% of products generated for this proxy study were formed with comparable reactivities (see Figure S19) using amines from either MBBC or Enamine, thereby providing confidence for both the identity and quality of amines obtained from MBBC. For each aryl halide used in Part II (Figure 4B), a Suzuki–Miyaura coupling reaction was conducted.⁵² The presence of Suzuki–Miyaura coupling product was confirmed for >80% of the aryl bromides and taken to be a positive indicator of the identity of the aryl bromide (see Figure S20).

Assessment of Reaction Reproducibility. Variance due to minor differences in the reaction setup performed on different days or reaction plates was also studied. To assess this variance, reactions that were repeated across different reaction plates were evaluated using regression and classification analyses (Figure 5).

Regression analysis entailed graphing reaction outcomes (as product LCAPs) from Part I against the product LCAPs from Part II. As shown in Figure 5B, the absolute reproducibility using regression analysis is suboptimal for a number of products ($R^2 = 0.35$). In contrast to the regression analysis, a binary classification analysis using a product LCAP threshold of 20% is highly consistent for most of the repeated reactions. This is reflected by the low number of false positives (FPs) (1) and false negatives (FNs) (4) for these repeated reactions. These results suggest that the classification analysis is a more robust strategy to account for inherent experimental noise than the corresponding regression study for our dataset.

Modeling. We next used our structurally diverse datasets to build and evaluate predictive models. As noted above, classification models are more suited to our HTE dataset and hence were the focus of this work (see the SI for regression modeling results). Literature reports suggest that models trained on random splits of HTE data generally perform well within the modeled datasets,^{11,16,29,30,34,35} an observation that is hypothesized to be a result of hidden patterns in the dataset or bias in its construction.⁵³ However, extending models to unseen structures is often difficult and limited by narrow substrate scopes.⁵⁴ In this study, the generation of a large, diverse dataset using 357 aryl halides and 348 amines enabled the arrangement of the train/validation/test splits in various ways to systematically interrogate the extrapolative and interpolative performance of the

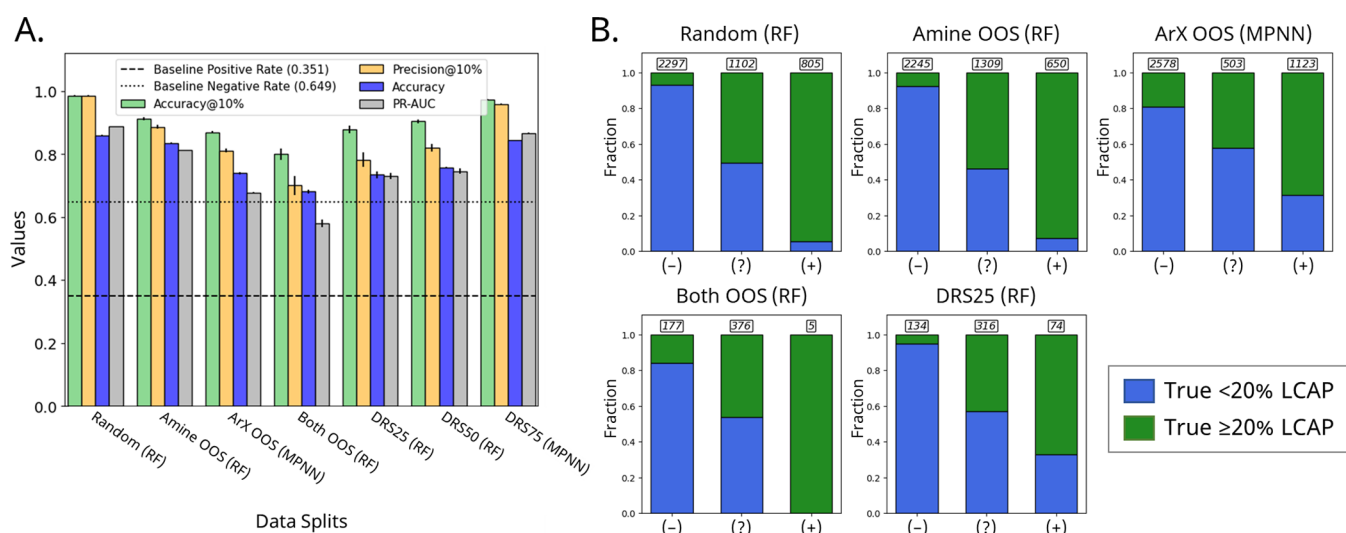


Figure 7. Categorical modeling results. (A) Accuracy@10%, precision@10%, accuracy, and PRAUC of the best models (RF = Random Forest, MPNN = message-passing neural network) for each data split. The values are given as averages across the N -fold validation (see SI for details) and error bars represent ± 1 standard error from three runs of training and inference. (B) Predictions binned by model confidence. (+), (−), (?) indicate predictions of $\geq 20\%$ LCAP with high confidence (threshold 0.7), $< 20\%$ LCAP with high confidence (threshold 0.7), and ambiguous predictions, respectively. Rounded average number of predictions from three runs is denoted in the textbox for each bin. See SI for the definition of model confidence.

models. In addition to the commonly used random split (Figure 6-I), two different paradigms for splitting the data were investigated.

The first paradigm features three different splits to interrogate the extrapolative model performance: Amine out-of-scope (OOS), ArX OOS, and Both OOS (Figure 6-II–IV, respectively). The test set for the Amine OOS model contains reactions between aryl halides in the training set and unseen amines. Similarly, the test set for the ArX OOS model consists of couplings of amines in the training set with unseen aryl halides. The Both OOS scenario interrogates the test set containing data points where both coupling partners are unseen, thereby evaluating the model's ability to fully extrapolate to unseen amine and aryl halide chemical space.

The second paradigm uses the “dimensionality reduction split” (DRS, Figure 6-V) to test the interpolative performance of the model by prediction in the n^2 space. Specifically, the model predicts the performance of couplings for which both the aryl halide and the amine components have been seen by the model but not in the combination present in the test set. Notably, if the modeling using a DRS is successful, this would imply that predictions for a full factorial space could be achieved using training sets that are a fraction of the n^2 space, thereby significantly reducing the experimental demands.

Categorical Modeling of Reaction Space. Figure 7 reports modeling results for random, DRS, and OOS splitting strategies. The data were modeled by treating LCAP as a categorical variable split into two groups using a product LCAP threshold of 20% (see the SI for 0% product LCAP classification results).⁵⁵ The resulting binary classification task enables the identification of sufficiently productive reactions ($\geq 20\%$ product LCAP) for medicinal chemistry campaigns.⁵⁶ Random Forest (RF), Decision Tree, AdaBoost, fully connected neural network, and message-passing neural network (MPNN)^{57–59} model architectures were investigated using one-hot encodings, Morgan fingerprints, quantum mechanical (QM) fingerprints (from descriptors calculated using a trained ML model),⁶⁰ molecular graphs, or some combination thereof as model inputs. Each classification model was evaluated based on four metrics:

accuracy, precision@10%, accuracy@10%, and precision-recall area under curve (PRAUC). Accuracy is defined as the percentage of reactions that are classified correctly in the appropriate product LCAP bin. Precision@10% refers to the prediction accuracy of the top 10% of reactions that the model most confidently predicts to have $\geq 20\%$ LCAP. Accuracy@10% is the average of precision@10% and the prediction accuracy of the top 10% of reactions that the model most confidently predicts to have $< 20\%$ LCAP. Finally, PRAUC is the area under the precision-recall curve that represents the robustness of a classifier. PRAUC ranges between 0 and 1, with higher values indicating more robust classifiers.

Figure 7A shows the results of the best model for each splitting strategy (see the SI for full details including model input features and hyperparameters). Best-performing model types, input features, and hyperparameters were determined based on accuracy@10% because this metric is a composite of the prediction accuracy for both low- and high-yielding reactions with high confidence (see SI for details). RF or MPNN classifiers exhibited the highest performance among all explored model types for all splitting strategies. Supplementing Morgan fingerprints with QM fingerprints was beneficial, depending on the type of data split and the model architecture (see Figure S73). Numbers in DRS notations represent the percentage of reactions used in the training set. For example, DRS25 uses 25% of reactions in the dataset as the training set (i.e., 75% reduction in data). The baseline positive rate (PR) for $\geq 20\%$ LCAP threshold is 35.1%, which reflects the percentage of data points in the dataset that exceed $\geq 20\%$ LCAP. This PR is the maximum possible hit rate for our dataset (i.e., % of reactions selected with LCAP $\geq 20\%$) without using a model. The baseline negative rate (NR) for $< 20\%$ LCAP threshold is 64.9%.

Accuracy@10% was high across all splits, confirming that the model confidence (see the SI for the definition of model confidence based on the model score) can be used to identify promising reactions or discard reactions that are highly unlikely to succeed (Figure 7A). Overall, $\geq 68\%$ accuracy and $\geq 80\%$ accuracy@10% demonstrates high predictivity of these models for all splits. In general, precision@10% is lower than accuracy@

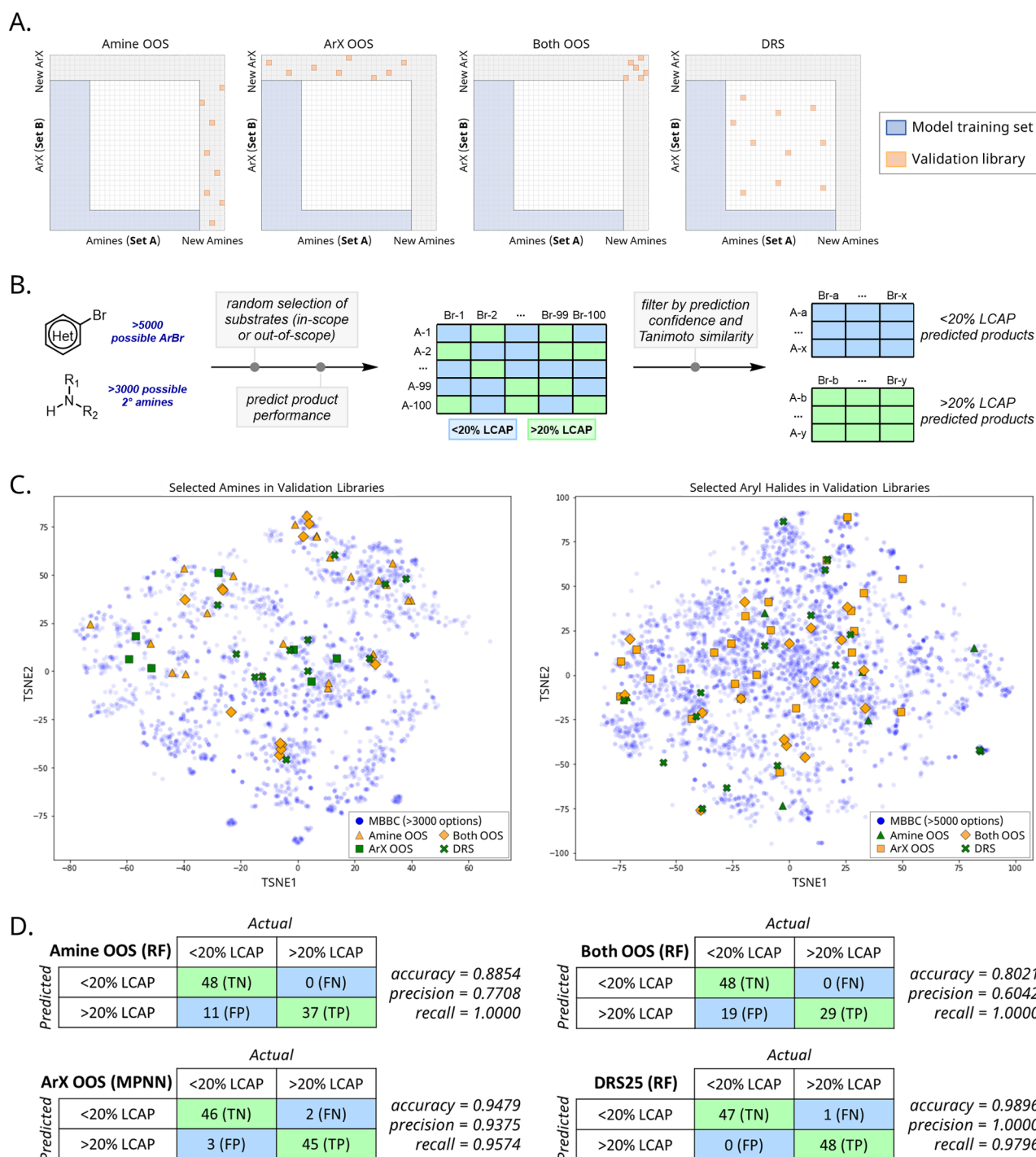


Figure 8. Validation library analyses. (A) Library design for each different data split with training data in light blue and new experimental validation data in light orange. (B) Workflow for the selection of amines and aryl halides for the validation libraries. Models were trained on the whole experimental dataset with the best-performing architecture, input feature, and hyperparameter combinations identified in Figure 7 (also see Figure S74). Model settings for DRS25 were used for the DRS validation library analysis. A prediction model confidence threshold of >0.9 was used for Amine OOS, ArX OOS, and DRS libraries, and >0.8 threshold was used for Both OOS library. For OOS amines or aryl halides, Tanimoto similarity was calculated against scope amines or aryl halides and a max similarity threshold of <0.7 was used. (C) Selected amines (left) and aryl halides (right) for validation libraries (blue circles = MBBC chemical space; triangles = Amine OOS library substrates; squares = ArX OOS library substrates; diamonds = Both OOS library substrates; crosses = DRS library substrates; orange markers = new substrates; green markers = scope substrates used in the dataset for the model build). (D) Confusion matrices with the results of the validation libraries (RF = Random Forest; MPNN = message-passing neural network; TN = true negative; FN = false negative; FP = false positive; TP = true positive).

10% implying that the models are better at predicting <20% LCAP reactions, which is likely a result of the skewed distribution of the dataset (35.1% PR). This is further illustrated in Figure 7B in which the predictions are binned into three

categories based on model confidence: predictions with high model confidence (>0.7) for either <20% LCAP (−) or ≥20% LCAP (+), as well as ambiguous predictions (?), which reflect model confidence <0.7 for either low- or high-yielding reactions.

The number of confident $\geq 20\%$ LCAP predictions is generally smaller than the number of predictions in the remaining two binned categories for most splits, which is consistent with the aforementioned model's lower predictive performance for high-yielding reactions. Importantly, the lowest calculated precision@10% for $\geq 20\%$ product LCAP threshold of Both OOS split ($>70\%$) exceeds PR ($\sim 35\%$) by 2-fold. This precision@10%:PR of 2 implies that the model can enhance the percentage of hits ($\geq 20\%$ product LCAP) by 100%, even with completely new substrates.

While the overall prediction accuracy of Both OOS split may appear to be comparable to NR, this is primarily due to the lower accuracy on high-yielding reactions. As shown in the integral accuracy averaging plot in Figure S77, prediction accuracies for $\geq 20\%$ and $<20\%$ LCAP reactions exceed PR and NR, respectively, confirming the model's robust predictive performance. This is further supported by the strong correlation between model confidence and accuracy in Figure S77, as well as the dummy classifier comparison in Table S2.

The models' capability to interpolate can be evaluated using two comparisons: random split and DRS vs OOS splits, and DRS25 vs DRS50 vs DRS75. Random split and DRS predictions are more accurate than Amine OOS, ArX OOS, and Both OOS predictions, showing that extrapolation is more challenging than interpolation. Furthermore, increasing prediction accuracy from DRS25 to DRS50 and DRS75 demonstrates that the models benefit from larger training sets. Nonetheless, it is exciting that the model can interpolate effectively with 75% reduction in data (DRS25), implicating the possibility of building reliable predictive models with smaller datasets. Moreover, the high predictivity for the DRS split indicates that a full factorial dataset (n^2 space) is not required to construct a highly interpolative model.

The models' extrapolative robustness is exhibited by the high predictivity for Both OOS split (accuracy $\geq 68\%$; accuracy@10% $\geq 80\%$) suggesting that the model learned the general reactivity patterns in the dataset. As expected, however, Both OOS predictions were less accurate compared with the predictions for Amine OOS and ArX OOS splits. A noteworthy observation is that the Amine OOS predictions were more accurate than the ArX OOS predictions. This could in part be attributed to the observed reactivity trends being more similar across different amines than across different ArX (see Figures S54–S56 and related section in the SI).

Last, we note that the model build workflow discussed above presents a challenge regarding feature importance and model interpretability. Since the model architecture and input features were optimized separately for each splitting strategy, it is difficult to identify generalizable trends across all model architectures. Moreover, interpreting important features for each model when using fingerprint vectors is complicated as the elements of fingerprint vectors may not necessarily correlate with intuitive physical or chemical quantities.

With the models in hand, we next designed 96 membered libraries for each split (except random split) to validate the model performance (Figure 8A). The workflow for the selection of amines and halides for these libraries is illustrated in Figure 8B. Amines and aryl bromides were randomly selected from the possible chemical space such that they fulfilled the criterion for each split. For each library, 50% of the products were predicted to have $>20\%$ LCAP with 80–90% model confidence, while the remaining 50% were predicted to have $<20\%$ LCAP. In addition, the Tanimoto similarity for OOS amines or aryl halides was <0.7

to those used in the dataset for the model build.⁶¹ Notably, the amines and halides selected for these validation libraries spanned the chemical space well (Figure 8C, orange selections). As shown in Figure 8D, the accuracy for these libraries was $>80\%$ across all splits (see Figures S25–S52 and related section in the SI for full details). The predictivity is higher for the ArX OOS and DRS splits than the Amine OOS and Both OOS splits.⁶² There are either zero or few FNs for these libraries, which has important implications since FN could lead to wasted time and resources on reaction screening for successful cross-couplings. The lower accuracy of the Amine OOS and Both OOS splits largely arises due to FPs, which could lead to the use of precious substrates for negative outcomes. The higher percentage of FP versus FN is consistent with the model being less confident in predicting higher yielding reactions as stated above (Figure 7B), resulting in more incorrect $>20\%$ predictions.

CONCLUSIONS AND OUTLOOK

This report details the construction of ML models for Pd-catalyzed C–N couplings using a large, structurally diverse, and pharmaceutically relevant HTE dataset. This is a significant advance as it offers a complementary strategy to prior reports on C–N coupling ML models that use HTE datasets with very narrow structural diversity.^{16,21} In all, this work is an important contribution to a relatively nascent field of building ML models for pharmaceutically relevant cross-couplings. The dataset generation was enabled by the discovery of novel nanomole scale compatible automation friendly C–N coupling reaction conditions. Specifically, LiOTMS was identified as an optimal base, and its use was previously unknown for C–N couplings.⁴⁷ The structural diversity in the dataset was achieved by leveraging MBBC. The large dataset enabled the systematic evaluation of model performance using five different data-splitting strategies, which were carefully designed to evaluate the model's ability to interpolate and extrapolate. Classification models were built with a lens toward application in medicinal chemistry campaigns. The models exhibit high predictive performance across all splits as gauged by standard metrics. Furthermore, model performance was experimentally interrogated using validation libraries that demonstrated high accuracy across all splits. Particularly exciting is the high accuracy obtained for Both OOS split, demonstrating the extrapolative robustness. In addition, the high accuracy with the DRS split implicates that predictions on a full factorial space can be achieved using a fraction of n^2 dataset. Taken together, these results would manifest as a significant enrichment of successful C–N coupling reactions by using the hits recommended by the models. In a prospective application, this would enable *in silico* identification of reaction performance with high fidelity, thereby focusing the time-consuming screening efforts on low-yielding targets. Ultimately, this paper lays the groundwork for building predictive models for structurally diverse large HTE datasets. Importantly, this workflow is not restricted to C–N coupling but could be applied to modeling efforts for other reaction classes. Such efforts have enormous potential to accelerate and enhance the resource efficiency of drug discovery efforts. Ongoing efforts seek to investigate alternative dataset design strategies and will be reported in due course.

ASSOCIATED CONTENT

Data Availability Statement

The experimental reaction dataset and all ML codes are available in the Figshare repository under the following DOI: 10.6084/

m9.figshare.28215923. The Pistachio patent dataset is a property of NextMove Software and can be obtained directly from NextMove Software, subject to the acquisition of a valid license (<https://www.nextmovesoftware.com/>).

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.5c00933>.

Details on HTE optimization and procedures, selection of secondary amines and aryl halides for scope experiments, procedures for amine and aryl halide quality assessment, substrate selection and nanoscale procedures for validation libraries, product scale-up and characterization including NMR spectra, and modeling details including featurization, data curation, data splitting, and additional results (PDF)

AUTHOR INFORMATION

Corresponding Authors

Dipannita Kalyani – Discovery Chemistry, Merck & Co., Inc., Rahway, New Jersey 07065, United States; orcid.org/0000-0003-4349-8016; Email: dipannita.kalyani@merck.com

Klavs F. Jensen – Department of Chemical Engineering, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139, United States; orcid.org/0000-0001-7192-580X; Email: kfjensen@mit.edu

Authors

Seung Kyun Ha – Department of Chemical Engineering, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139, United States; orcid.org/0000-0003-2967-1097

Michael S. West – Discovery Chemistry, Merck & Co., Inc., Rahway, New Jersey 07065, United States

Jessica Xu – Department of Chemical Engineering and Department of Chemistry, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139, United States; orcid.org/0000-0003-2587-9587

Yu-hong Lam – Discovery Chemistry, Merck & Co., Inc., Rahway, New Jersey 07065, United States; orcid.org/0000-0002-4946-1487

Thomas Struble – Discovery Chemistry, Merck & Co., Inc., Rahway, New Jersey 07065, United States

Spencer Dreher – Discovery Chemistry, Merck & Co., Inc., Rahway, New Jersey 07065, United States; orcid.org/0000-0002-5094-1218

Shane W. Krska – Discovery Chemistry, Merck & Co., Inc., Rahway, New Jersey 07065, United States; orcid.org/0000-0001-9757-9036

Stephen L. Buchwald – Department of Chemistry, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139, United States; orcid.org/0000-0003-3875-4775

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/jacs.5c00933>

Author Contributions

[#]S.K.H., D.K., M.S.W., and J.X. contributed equally to this work.

Notes

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- (46) In Figure 3A Step 2, the goal of the base screen was to identify reagents that are widely well-performing. Informer X13 is known to be base sensitive, thus Informer X2 was used in this step instead to ensure the identification of high-performing bases for general substrates.
- (47) After the preliminary disclosure of this work ((a) Xu, J.; Kalyani, D.; Struble, T.; Dreher, S.; Krska, S.; Buchwald, S. L.; Jensen, K. F. Roadmap to Pharmaceutically Relevant Reactivity Models Leveraging High-Throughput Experimentation. *ChemRxiv* **2022**. This content is a preprint and has not been peer-reviewed), the Buchwald group published a report on C–N couplings using NaOTMS as a base:

(b) Reichert, E. C.; Feng, K.; Sather, A. C.; Buchwald, S. L. Pd-Catalyzed Amination of Base-Sensitive Five-Membered Heteroaryl Halides with Aliphatic Amines. *J. Am. Chem. Soc.* **2023**, *145*, 3323–3329. We have also observed comparable performance using NaOTMS and slightly lower performance using KOTMS.

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(62) The higher accuracy observed for the ArX OOS library vs the Amine OOS library is contrary to the results discussed in [Figure 7A](#). This result could possibly be explained by the small sample size of the validation library or that accuracy could fluctuate depending on the species used for validation. Moreover, the number of FPs in the Amine OOS library is higher than expected due to the model incorrectly predicting the coupling results of an individual amine for every reaction (see [SI](#) for more details), which ultimately lowers the calculated accuracy.



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